LAB ASSIGNMENT-3

DDL Questions (Data Definition Language)

1. Create the TEACHER table with the following attributes:

(Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).

```
CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary INT
);
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
Name	varchar(50)	YES		NULL	
DeptNo	int	YES		NULL	
Date_of_Joining	date	YES		NULL	
DeptName	varchar(50)	YES		NULL	
Location	varchar(50)	YES		NULL	

2. Create the SALES table with the following attributes:

(Sales_No, Salesname, Branch, Salesamount, DOB).

```
CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount INT,
    DOB DATE
);
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
▶	Sales_No	int	NO	PRI	NULL	
	Salesname	varchar(50)	YES		NULL	
	Branch	varchar(50)	YES		NULL	
	Salesamount	int	YES		NULL	
	DOB	date	YES		NULL	

3. Create the DEPARTMENTS table with the following attributes: (dept_no, dept_name, dept_location).

```
CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
```

OUTPUT:

	Field	Type	Null	Key	Default	Extra
▶	dept_no	int	NO	PRI	NULL	
	dept_name	varchar(50)	YES		NULL	
	dept_location	varchar(50)	YES		NULL	

4. Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```sql
CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary INT,
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

OUTPUT:

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(50)	YES		NULL	
emp_salary	int	YES		NULL	
dept_no	int	YES	MUL	NULL	

DML Questions (Data Manipulation Language)

From TEACHER Table

1.Insert five records into the TEACHER table.

```sql
INSERT INTO TEACHER VALUES
('Nakul Arora', 10, '2016-06-15', 'Mathematics', 'Delhi', 40000),
('Vansh Tanwar', 20, '2017-07-10', 'Commerce', 'Mumbai', 45000),
('Rahul Kumar', 10, '2015-05-20', 'Mathematics', 'Delhi', 42000),
('Meena Patel', 30, '2018-08-25', 'Physics', 'Ahmedabad', 48000),
('Kiran Das', 20, '2019-09-12', 'Commerce', 'Kolkata', 46000);
```

OUTPUT:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Nakul Arora	10	2016-06-15	Mathematics	Delhi	40000
Vansh Tanwar	20	2017-07-10	Commerce	Mumbai	45000
Rahul Kumar	10	2015-05-20	Mathematics	Delhi	42000
Meena Patel	30	2018-08-25	Physics	Ahmedabad	48000
Kiran Das	20	2019-09-12	Commerce	Kolkata	46000

2.Give an increment of 25% salary to teachers working in the Mathematics department.

```
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

OUTPUT:

Name	DeptName	Salary
Nakul Arora	Mathematics	50000
Rahul Kumar	Mathematics	52500

3.Give an increment of 15% salary to teachers working in the Commerce department.

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

OUTPUT:

Name	DeptName	Salary
Sunita Rao	Commerce	51750
Kiran Das	Commerce	52900

From SALES Table

1.Insert five records into the SALES table.

```
INSERT INTO SALES VALUES
(201, 'Amit',   'Delhi',     120000, '1996-12-21'),
(202, 'Neha',   'Mumbai',     95000, '1995-07-14'),
(203, 'Rohit',  'Bangalore', 150000, '1994-12-09'),
(204, 'Sneha',  'Delhi',      87000, '1997-03-18'),
(205, 'Karan',  'Mumbai',    110000, '1993-12-25');
```

OUTPUT:

Sales_No	Salesname	Branch	Salesamount	DOB
201	Amit	Delhi	120000	1996-12-21
202	Neha	Mumbai	95000	1995-07-14
203	Rohit	Bangalore	150000	1994-12-09
204	Sneha	Delhi	87000	1997-03-18
205	Karan	Mumbai	110000	1993-12-25

2.Calculate the total sales amount in each branch.

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT:

Branch	Total_Sales
Delhi	207000
Mumbai	205000
Bangalore	150000

3.Calculate the average sales amount in each branch.

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

OUTPUT:

	Branch	Average_Sales
▶	Delhi	103500.0000
	Mumbai	102500.0000
	Bangalore	150000.0000

4.Display the Salesname and DOB of salesmen born in the month of December with date format like 21-Dec-09.

```
SELECT Salesname,
       DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

OUTPUT:

	Salesname	DOB
▶	Amit	21-Dec-96
	Rohit	09-Dec-94
	Karan	25-Dec-93

5.Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
SELECT Salesname,
       DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

OUTPUT:

Salesname	DOB
Amit	21-Dec-96
Rohit	09-Dec-94
Karan	25-Dec-93
Neha	14-Jul-95
Sneha	18-Mar-97

From JOIN Operations

1.Display employee details and department details where the department numbers are the same in both tables.

```
SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;
```

OUTPUT:

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
1	Amit	50000	10	10	HR	Delhi
3	Rahul	45000	10	10	HR	Delhi
2	Neha	60000	20	20	IT	Bangalore
4	Sneha	70000	20	20	IT	Bangalore
5	Karan	40000	30	30	Sales	Mumbai

2.Display the employee name and department name using a LEFT OUTER JOIN.

```sql
SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
LEFT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;
```

OUTPUT:

emp_name	dept_name
Amit	HR
Neha	IT
Rahul	HR
Sneha	IT
Karan	Sales

3.Display the employee name and department name using a RIGHT OUTER JOIN.

```sql
SELECT e.emp_name, d.dept_name
FROM EMPLOYEES e
RIGHT JOIN DEPARTMENTS d
ON e.dept_no = d.dept_no;
```

OUTPUT:

emp_name	dept_name
Amit	HR
Rahul	HR
Neha	IT
Sneha	IT
Karan	Sales

4.Display the details of employees who draw a salary greater than the average salary.

```sql
SELECT emp_id, emp_name, emp_salary
FROM EMPLOYEES
WHERE emp_salary >
      (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

OUTPUT:

emp_id	emp_name	emp_salary
2	Neha	60000
4	Sneha	70000
6	Vikas	55000
NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1.Perform the ROLLBACK command on the TEACHER table.

```sql
START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 2000;
ROLLBACK;
```

OUTPUT:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anil Kumar	10	2016-06-15	Mathematics	Delhi	40000
Sunita Rao	20	2017-07-10	Commerce	Mumbai	45000
Ravi Sharma	10	2015-05-20	Mathematics	Delhi	42000
Meena Patel	30	2018-08-25	Physics	Ahmedabad	48000
Kiran Das	20	2019-09-12	Commerce	Kolkata	46000

2.Perform the COMMIT command on the TEACHER table.

```
START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 3000;
COMMIT;
```

OUTPUT:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anil Kumar	10	2016-06-15	Mathematics	Delhi	43000
Sunita Rao	20	2017-07-10	Commerce	Mumbai	48000
Ravi Sharma	10	2015-05-20	Mathematics	Delhi	45000
Meena Patel	30	2018-08-25	Physics	Ahmedabad	51000
Kiran Das	20	2019-09-12	Commerce	Kolkata	49000

3.Write a query to implement a SAVEPOINT.

```
START TRANSACTION;
UPDATE TEACHER
SET Salary = Salary + 1000;
SAVEPOINT A;
UPDATE TEACHER
SET Salary = Salary + 500;
ROLLBACK TO A;
COMMIT;
```

OUTPUT:

Name	DeptNo	Date_of_Joining	DeptName	Location	Salary
Anil Kumar	10	2016-06-15	Mathematics	Delhi	44000
Sunita Rao	20	2017-07-10	Commerce	Mumbai	49000
Ravi Sharma	10	2015-05-20	Mathematics	Delhi	46000
Meena Patel	30	2018-08-25	Physics	Ahmedabad	52000
Kiran Das	20	2019-09-12	Commerce	Kolkata	50000

DCL Questions (Data Control Language)

1.Write a query to grant all privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT ALL
ON EMPLOYEES
TO DEPARTMENTS;
```

2.Write a query to grant selected privileges on the EMPLOYEES table to DEPARTMENTS.

```
GRANT SELECT, INSERT
ON EMPLOYEES
TO DEPARTMENTS;
```

3.Write a query to revoke all privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL
ON EMPLOYEES
FROM DEPARTMENTS;
```

4.Write a query to revoke selected privileges on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE SELECT, INSERT
ON EMPLOYEES
FROM DEPARTMENTS;
```